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A unified approach for relationships among Green's function, normal modes and dispersion spectrum in layered elastic half-space, with corrected misconceptions on surface wave dispersion and testing

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SUMMARY

In this paper, the global stiffness matrix [***K***] and the Fourier–Bessel series methods are proposed to derive the accurate Green's function and dynamic response in a form that is directly related to the dispersion curve and experimental dispersion spectrum. Detailed analyses are carried out for the two-layered half-space with different velocity profiles, including the homogeneous half-space as a special case. Our studies indicate that, in Rayleigh wave analysis, the original Rayleigh equation, instead of the rationalized Rayleigh equation as previously derived and used, should be used since the latter would contain extra non-physical roots. We further reveal and characterize three distinct types of leaky waves: the intrinsic surface leaky wave, the apparent Rayleigh mode with a frequency gap associated with a low-velocity half-space and the fast-guided *P–SV* wave in the layered medium with a high V_S contrast between the upper layer and the lower half-space. All leaky modes can be captured by local minima of |det[***K***]| instead of tracing complex roots in other existing approaches. In the experimental estimation of dispersion curves for practical applications, we have observed that the truncation effect is the major source of uncertainty regardless of the wavefield transformation method utilized. Furthermore, the truncation effect is both location- and model-dependent, without a unique optimal near offset. As such, in order to reduce the uncertainty from the truncation effect, the receiver layout should be considered in the inversion of dynamic response, instead of relying on ensuring a minimum near offset. This becomes possible with the present fast and accurate complete dynamic Green's function by which all wave phenomena (including different types of leaky waves) and receiver locations can be considered in the wavefield transformation.

Key words: Guided waves; Surface waves and free oscillations; Wave propagation.

1 INTRODUCTION

Seismic waves are most commonly generated by vertical impacts and recorded on the ground surface in a shallow seismic testing. In a homogeneous half-space under such loading, a type of surface wave known as Rayleigh waves is guided along the near surface of solids in addition to the radiated body waves. They are dispersive when guided in layered half-space and their dispersive characteristics have provided valuable information for non-destructive measurements of shear (*S*)-wave velocity. Various approaches were proposed for seismic testing based on the surface wave, such as the well-known spectral analysis of surface waves (SASW) method (Nazarian 1984) and multichannel analysis of surface waves (MASW) method (Park *et al.* 1999; Xia *et al.* 1999). While the algorithm may differ, all methods involve the computation of Rayleigh-wave dispersion curves (i.e. phase velocities for a given frequency), experimental estimation of dispersion curves and inversion based on dispersion curve fitting. Hence, correct modelling of dispersion curves and their accurate determination from seismic testing data are crucial in any surface wave survey.

In the forward computation of dispersion curves (modal solutions), the convenient transmission matrix method (also known as the Thomson–Haskell method) was first presented by Thomson (1950) and later corrected by Haskell (1953). Since then, many methods have been proposed to improve the numerical stability and computation efficiency, including the δ matrix method (Dunkin 1965; Thrower 1965), Schwab–Knopoff method (Schwab & Knopoff 1970), Abo-Zena method (Abo-Zena 1979; Zhang *et al.* 1997), global matrix method (Knopoff 1964; Randall 1967) and reflection/transmission (R/T) coefficients method (Kennett 1974; Chen 1993). These methods work well under normal conditions where the *S*-wave velocity (V_S) of the lower half-space is the highest. However, when the *S*-wave velocity of the underlying half-space is lower than some of the layers above (typical situation for a pavement

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profile), real roots cannot be found at some frequencies (Roësset 1998; Rydén & Park 2004; Pan *et al.* 2013). Modal solutions in terms of plane waves are unable to describe leaky waves that emit energy in the form of outgoing waves into the half-space and decay in amplitude as they travel. In this case, an attenuation term needs to be introduced even if all the materials are purely elastic. This can be achieved by introducing complex frequency or complex wavenumber (Lowe 1995), the latter being more common. However, tracing the roots of the modal characteristic function in the complex domain is substantially more difficult than for the real roots.

In seismological applications, near surface is often covered by unconsolidated sediments with high Poisson's ratios. Another type of leaky waves (also called guided waves) has been observed, which are associated with a strong velocity contrast and high Poisson's ratio (Roth *et al.* 1998; Roth & Holliger 2000; Gao *et al.* 2014; García-Jerez & Sánchez-Sesma 2015; Wu & Chen 2017; Sun *et al.* 2021). As the phase velocity becomes higher than the maximum V_S, the roots of the modal characteristic function become complex, which correspond to the leaking mode (Gao *et al.* 2014; Sun *et al.* 2021). This kind of leaky wave propagates faster than the Rayleigh wave and attenuates along the direction of propagation. The presence of leaky waves may either obstruct correct identification of normal modes, hence affecting the accurate inversion of the *S*-wave velocity (Gao *et al.* 2014), or provide useful information for characterizing velocity structures (Boiero *et al.* 2013). Both synthetic and real-world examples have been given to demonstrate significant overestimation of *S*-wave velocity due to mode misidentification resulted from the leaky wave (Gao *et al.* 2014, 2016). Compared with normal modes, leaky modes are not well understood and rarely utilized in near-surface investigation due to their complexity and difficult distinction from normal modes. The key and difficult point of using guided wave in the inversion is mode identification (Boiero *et al.* 2013). While the Rayleigh waves are formed by the superposition of normal modes, guided waves are considered to be formed by the superposition and interference of multiple reflections of *P* and *SV* waves in the waveguides, and it is dispersive. Sun *et al.* (2021) used the transmission matrix approach to compute the complete theoretical dispersion curves (normal modes and leaking modes) of *P–SV*-wave and pure *P*-wave systems for two-layered models with a higher-velocity lower half-space. The leaking modes are divided into two regions, non-geometric $\bar{\mathrm{P}}S$ wave (*P* component is inhomogeneous and *S* component is homogeneous) in $V_{S2} < V < V_{P1}$ and residual guided wave in $V_{P1} < V < V_{P2}$. In an effort to better characterize guided waves, they also defined a relative amplitude dispersion map (RADM) to include the polarized energy ratio (vertical to horizontal) information on the dispersion curves. A leaky surface wave has also been observed at the surface of an isotropic homogeneous elastic solid. Schröder & Scott (2001) theoretically explained this as a result of the complex conjugate roots of the Rayleigh equation and showed it exists only for materials with a Poisson's ratio greater than about 0.263. However, there are no multiple reflections of *P* and *SV* waves in any homogeneous half-space, the mechanism the leaky surface wave in homogeneous half-space should be different from the guided waves in multi-layered media. So far, this has not been clearly distinguished from the guided waves in the literature.

It can be seen from the above literature reviews that information regarding the leaky waves are quite fragmented. A unified approach is yet to be presented to explain and model all phenomena in surface wave dispersion. It should also be noted that, while modal solutions (real roots for normal modes or complex roots for leaky modes) reveal all possible combinations of phase velocity and frequency, they do not give any information on which portions of different modes would be measurable at the surface. Amplitude ratios between modes have been derived to calculate the apparent or effective dispersion curve where the experimental set-up is taken into account (Tokimatsu *et al*. 1992; Lai 1998). Rydén & Park (2004) examined the excitability of a particular mode at a particular frequency, which is defined as the ratio of displacement of that mode to the applied force (Wilcox *et al.* 2002), independent of receiver locations. To this end, a more quantitative and direct indication of the excitability should be the Green's function. Green's functions are in the middle step of generating synthetic seismograms (Schmidt & Tango 1986; Pan 2019; Lin *et al.* 2022). While modal solutions in terms of dispersion curves and response solutions in terms of synthetic seismograms are frequently presented in surface wave modelling, Green's functions are rarely examined. Lin *et al.* (2022) recently presented a very efficient approach based on the global stiffness matrix and the Fourier–Bessel series methods to compute the Green's functions and dynamic response in layered media. It can be used for efficiently comparing the modal solutions with the Green's functions to provide more insights to various leaky modes. Finally in the aspect of experimental estimation of the dispersion curve or Green's function from field data, many wavefield transformation algorithms have been proposed. They can be roughly divided into two major groups, plane wavefield transformation methods (e.g. McMechan & Yedlin 1981; Yilmaz 1987; Park *et al.* 1998; Lin & Chang 2004; Xia *et al.* 2007) and cylindrical wavefield transformation methods (e.g. Forbriger 2003; Zywicki & Rix 2005; Wang *et al.* 2019; Xi *et al.* 2021). The results of these wavefield transformation methods are often presented as dispersion images in terms of the frequency–velocity spectrum (FVS). The theoretical Green's function should be compared with the experimental FVS to understand what are actually observable as a function of the field set-up and transformation algorithm.

In this study, the modal solution, Green's function and dynamic response are all derived in terms of the global stiffness matrix to establish the explicit and direct links among them. Based on such a unified approach, the theoretical Green's function is connected to the dispersion curves (including normal modes and leaky modes) and observable field quantities. We first examine the characteristics of theoretical Green's function in different types of earth models. We then connect them to the dispersion curves. Finally, we compare the Green's function with the observed dispersion spectrum (FVS). The results reveal some existing misconceptions about leaky waves and experimental design. Implications of the unified approach to further improving surface wave inversion are also discussed.

2 METHODS

2.1 Theoretical dynamic response, Green function and dispersion curves

In a typical seismic testing, a dynamic loading (impact) is applied on the ground surface. Considering an axial symmetric, circular and vertical load and taking the circular centre as the origin of the radius coordinate, we then have, on the surface, the load expression as

$$P\ (t, r) = \ F\,(t)\,S\,(r)\,, \tag{1}$$

where t and r are the time and radius, and $F(t)$ and $S(r)$ are the time variation and spatial variation of the load, respectively. Through Fourier transform with respect to t and Hankel transform (Fourier–Bessel transform) with respect to r, the seismic source eq. (1) in the

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transformed domain is written as

$$\hat{P}\ (\omega, k) = \hat{F}\ (\omega)\, \hat{S}\,(k)\,, \tag{2}$$

where ω is frequency, k is wavenumber, $\hat{F}(\omega)$ is the Fourier transform of $F(t)$ and $\hat{S}(k)$ is the Hankel transform of $S(r)$. Note that $\hat{S}(k)$ is reduced to 1 for a concentrated point source.

The dynamic response to the applied load $P(t, r)$ can be expressed as a Fourier inverse transform of the frequency response (i.e. the response to time-harmonic loading):

$$w\ (t, r) = \int_{-\infty}^{\infty} W\,(\omega, r)\, \hat{F}\,(\omega)\, \mathrm{e}^{i\omega t}\, \mathrm{d}\omega \tag{3a}$$

$$u\ (t, r) = \int_{-\infty}^{\infty} U\,(\omega, r)\, \hat{F}\,(\omega)\, \mathrm{e}^{i\omega t}\, \mathrm{d}\omega, \tag{3b}$$

where $w(t, r)$ and u(t, r) are the vertical and radial displacement wavefields on the ground surface, which are of major interest in seismic testing; $W(\omega, r)$ and $U(\omega, r)$ are the corresponding frequency responses (also known as the Fourier coefficients of the displacements).

The frequency response can be derived as the inverse Hankel transform of the solution in the transform domain (Lin *et al.* 2022), namely,

$$W\ (\omega, r) = \int_{-\infty}^{\infty} \hat{W}\,(\omega, k)\, \hat{S}\,(k)\, J_0\,(kr)\, k\mathrm{d}k \tag{4a}$$

$$U\ (\omega, r) = \int_{-\infty}^{\infty} \hat{U}\,(\omega, k)\, \hat{S}\,(k)\, J_1\,(kr)\, k\mathrm{d}k, \tag{4b}$$

where $\hat{W}(\omega, k)$ and $\hat{U}(\omega, k)$ are the Green's functions of the vertical and radial displacements in the transform domain (simply called Green's functions hereafter); J_0 and J_1 are the Bessel functions of the first kind of orders 0 and 1, respectively. The kernel function (i.e. the Green's function $\hat{W}(\omega, k)$ or $\hat{U}(\omega, k)$) of a multilayered half-space can be derived by various approaches (Thomson 1950; Haskell 1953; Kennett 1983; Kausel & Roësset 1981; Müller 1985; Pan *et al.* 2021). In this study, the formulation based on the stiffness matrix method is adopted (Kausel & Roësset 1981; Lin *et al.* 2022). This approach can be thought as the finite-element method in the transformed domain and it is also called the spectral element method (Doyle 1997). For a multi-layered half-space subjected to a unit vertical load on the ground surface (in the Fourier and Hankel transformed domain), the interface displacements in the transformed domain are related to the interface forces (tractions) by the global stiffness matrix $[\boldsymbol{K}]$ as

$$[\boldsymbol{K}] \begin{bmatrix} \hat{U}\,(\omega, k) \\ \hat{W}\,(\omega, k) \\ \hat{U}_2\,(\omega, k) \\ \hat{W}_2\,(\omega, k) \\ \vdots \\ \vdots \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \tag{5}$$

in which, the Green's functions, $\hat{U}$ and $\hat{W}$, are the displacements on the ground surface, and the subscripts 2, ... in $\hat{U}$ and $\hat{W}$ denote those in the first layer interface, ...; The vector on the right-hand side is the interface force (traction) vector, which are all zero, except for the vertical load on the ground surface (equal to 1). Detailed derivation for the global stiffness matrix $[\boldsymbol{K}]$ is referred to Kausel & Roësset (1981) and Lin *et al.* (2022). By Cramer's rule, the surface displacements in the transformed domain, $\hat{U}$ and $\hat{W}$, can be expressed as

$$\hat{U}\ (\omega, k) = \frac{\det\,(\boldsymbol{K}_U)}{\det\,(\boldsymbol{K})} \tag{6a}$$

$$\hat{W}\ (\omega, k) = \frac{\det\,(\boldsymbol{K}_W)}{\det\,(\boldsymbol{K})} \tag{6b}$$

in which matrix $[\boldsymbol{K}_U]$ ($[\boldsymbol{K}_W]$) is formed by replacing the first (second) column of matrix $[\boldsymbol{K}]$ by the force vector on the right-hand side (i.e. eq. 5). For free vibrations, that is the right-hand side of eq. (5) becomes a zero vector, a non-trivial solution exists only when the following condition is satisfied:

$$\det\,[\boldsymbol{K}] = 0. \tag{7}$$

Equation (7) is the characteristic equation that leads to the normal mode dispersion relation. The dispersion points ($k = k_l\ (\omega)$, $l = 0, 1, 2, \ldots$) are the roots of the characteristic function (det$[\boldsymbol{K}]$). These dispersion points correspond to the singular points in the Green's function, namely, at which $\hat{W}(\omega, k)$ or $\hat{U}(\omega, k)$ tends to infinity. The dispersion points on each mode l constitute the dispersion curve (or branch). Within these curves, the fundamental mode, which corresponds to the lowest phase velocity $v = \omega/k$, is most important. Note that, in seismic engineering and for convenience, both the Green's functions and dispersion curves are also converted to the frequency–velocity domain as $\hat{W}(\omega, v)$ (*or* $\hat{U}(\omega, v)$) and $v = v_l\ (\omega)$. It is further noted that we derive the modal solution and response solution under the same framework without assuming the plane wave condition. Hence leaky waves can be accounted for without introducing the complex wavenumber or complex phase velocity.

While various formulations are available for finding the dispersion curves, the approach proposed here has certain advantages. For instance, the advantage of the stiffness matrix $[\boldsymbol{K}]$ in eq. (7) is that it is directly related to the Green's functions in eq. (6), in which det$[\boldsymbol{K}]$ is the denominator of the Green's functions. Consequently, the dispersion curves comprised of the singular points in det$[\boldsymbol{K}] = 0$ are associated with the energy peaks in the Green's functions. For a homogeneous half-space, as is well-known, the analytical dispersion equation, that is the Rayleigh equation, can be expressed as (e.g. Achenbach 1973)

$$\left(2 - \frac{v^2}{v_\mathrm{s}^2}\right)^2 + 4\sqrt{\frac{v^2}{v_\mathrm{p}^2} - 1}\,\sqrt{\frac{v^2}{v_\mathrm{s}^2} - 1} = 0. \tag{8}$$

In an effort to explain the observed leaky wave or non-geometric wave (Schröder & Scott 2001; Gao *et al.* 2014), eq. (8) is rationalized to

$$\left(\frac{v}{v_\mathrm{s}}\right)^2 \left[\left(\frac{v}{v_\mathrm{s}}\right)^6 - 8\left(\frac{v}{v_\mathrm{s}}\right)^4 + \left(24 - 16\left(v_\mathrm{s}/v_\mathrm{p}\right)^2\right)\left(\frac{v}{v_\mathrm{s}}\right)^2 - 16\left(1 - \left(v_\mathrm{s}/v_\mathrm{p}\right)^2\right)\right] = 0. \tag{9}$$

It should be pointed out that eq. (8) is the true Rayleigh Equation in the homogeneous half-space and it will be used to benchmark the dispersion curve in the layered half-space using eq. (7). Note that since the rationalized eq. (9) is obtained by multiplying a factor to eq. (8), extraneous roots may be introduced. Therefore, one should be extremely careful when using eq. (9) for the Rayleigh roots or dispersion curves. In other words, all the roots of eq. (9) need to be double checked, and correspondingly, explanation of the leaky

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wave or non-geometric wave should be redefined! These important issues will be addressed below via numerical examples.

Assuming finite interval and band-limited for the dynamic response, we can replace the Fourier transform with Fourier series and the Hankel transform with Fourier–Bessel series for efficient numerical computation. Therefore, the continuous frequency and wavenumber in the wavefield become discrete ω_n and k_m where a maximum time window T and number N determines the discrete points for ω_n, and a maximum radius R and number M determines the discrete points for k_m. The computational accuracy can be controlled by setting large enough values of R, M, T and N (Lin *et al.* 2022). Note that while ω_n are uniform discrete points, k_m are discrete points determined by the zeros of the involved Bessel function. In doing so, the dynamic response in eq. (3) becomes the Fourier series synthesis of $W(\omega_n, r)$ and $U(\omega_n, r)$, as

$$w\ (t, r) = \sum_n W\ (\omega_n, r)\ \hat{F}\ (\omega_n)\ \mathrm{e}^{i\omega_n t} \tag{10a}$$

$$u\ (t, r) = \sum_n U\ (\omega_n, r)\ \hat{F}\ (\omega_n)\ \mathrm{e}^{i\omega_n t}, \tag{10b}$$

and the frequency response in eq. (4) becomes the Fourier–Bessel series synthesis of $\hat{W}(\omega_n, k_m)\hat{S}(k_m)$ and $\hat{U}(\omega_n, k_m)\hat{S}(k_m)$

$$W\ (\omega_n, r) = \sum_m \hat{W}\ (\omega_n, k_m)\ \hat{S}\ (k_m)\ J_0\ (k_m r) \tag{11a}$$

$$U\ (\omega_n, r) = \sum_m \hat{U}\ (\omega_n, k_m)\ \hat{S}\ (k_m)\ J_1\ (k_m r)\,. \tag{11b}$$

Note that the left-hand sides in eq. (11) are the expansion coefficients on the right-hand sides of eq. (10). Thus, the dynamic response and frequency response are efficiently computed by eqs (10) and (11), respectively, instead of numerical integration for eqs (3) and (4).

2.2 Estimated Green's functions and dispersion curves from observed wavefield

The goal of seismic testing is to estimate the elastic properties of the multi-layered half-space from the observed seismic response. The wavefield is sampled both in the time and space domains, resulting in discrete wavefield, for instance for the vertical displacement $w(t_n, r_m)$. Due to the high-speed analogue to digital device, the sampling in time can be very dense with small time interval and sufficiently long-time window. The Nyquist frequency (half of the sampling frequency) can be higher than the highest possible frequency in the seismic record, with the time window being long enough to encompass the whole wave packet. Hence, there is basically no loss of information in the time domain. However, this is usually not the case for sampling in the space domain. Typically, only a limited number of geophones are available. The wavefield in the radial direction is loosely sampled at a fixed interval, that is Δr, and truncated within the geophone spread length.

If the full wavefield can be completely captured via seismic survey, the theoretical Green's function can be reconstructed. However, this is typically not the case in the space domain as we have just discussed. Therefore, we have to 'estimate' the Green's function and hence the dispersion curves, from the truncated data. This operation is often referred to as the wavefield transformation or dispersion analysis. Various algorithms have been proposed, such as the frequency–wavenumber (f–k) transform (i.e. Yilmaz 1987), slowness-frequency transform (i.e. McMechan & Yedlin 1981), phase-shift method (i.e. Park *et al.* 1998), frequency decomposition and slant stacking method (i.e. Xia *et al.* 2007). It can be shown that these methods are physically and mathematically equivalent (Lin *et al.* 2017). Therefore, we use the f–k transform in handling our observation data as discussed below.

Utilizing the f–k transform and converting the wavenumber variable to phase velocity, the following equation can be used to estimate the Green's function:

$$\hat{W}_{\mathrm{obs}}\ (\omega, v)\ = \sum_m W\ (\omega, r_m)\ \mathrm{e}^{i\frac{\omega}{v}r_m} \tag{12a}$$

$$\hat{U}_{\mathrm{obs}}\ (\omega, v)\ = \sum_m U\ (\omega, r_m)\ \mathrm{e}^{i\frac{\omega}{v}r_m}, \tag{12b}$$

where $\hat{W}_{\mathrm{obs}}(\omega, v)$ and $\hat{U}_{\mathrm{obs}}(\omega, v)$ are the estimated Green's functions for the vertical and radial components, which are called the frequency–velocity spectrum (FVS) hereafter; $W(\omega, r_m)$ and $U(\omega, r_m)$ are the Fourier transform of the corresponding observed wavefield $w(t_n, r_m)$ and $u(t_n, r_m)$. The dispersion curves are experimentally estimated by the amplitude peaks of the FVS. Note that the only difference between eq. (12) and the phase-shift method is that the amplitude of $W(\omega, r_m)$ or $U(\omega, r_m)$ is normalized to unity in the latter method. For convenience, eq. (12) is simply called the phase-shift method in this study. To estimate how the total energy is distributed over the wavenumber (phase velocity), eq. (12) is adopted to calculate the FVS instead of the amplitude-normalized phase-shift method or the linear Randon transform (Luo *et al.* 2009).

Comparing eq. (12) with eq. (4), it is clear that $\hat{W}_{\mathrm{obs}}(\omega, v)$ and $W(\omega, r)$ are not Fourier transform pairs, since eq. (12) will not approach $\hat{W}(\omega, v)$ even if very dense and long wavefield data are recorded. The counter part of eq. (4) should be

$$\hat{W}\ (\omega, v)\ \hat{S}\ (v) = \int_0^\infty W(\omega\ , r) J_0\left(\frac{\omega}{v}r\right) r\,\mathrm{d}r \tag{13a}$$

$$\hat{U}\ (\omega, v)\ \hat{S}\ (v) = \int_0^\infty U\omega\ , r) J_1\left(\frac{\omega}{v}r\right) r\,\mathrm{d}r. \tag{13b}$$

This means that the Fourier–Bessel transform of the frequency response is the theoretical Green's function multiplied by the source function $\hat{S}(v)$. $\hat{S}(v)$ approaches 1 when the source becomes a point load. Based on this notion, the Fourier–Bessel transform method was proposed for estimating the Green's function (Forbriger 2003; Wang *et al.* 2019; Xi *et al.* 2021). In practice, we do not know the continuous wavefield $W(\omega, r)$ or $U(\omega, r)$, but the discrete and truncated ones. Taking $\hat{S}(v) \cong 1$, eq. (12) is approximated by a trapezoidal rule as:

$$\hat{W}_{\mathrm{obs}}\ (\omega, v) = \sum_{m=1}^{M} W\ (\omega, r_m)\ J_0\left(\frac{\omega}{v}r_m\right) r_m \Delta r_m \tag{14a}$$

$$\hat{U}_{\mathrm{obs}}\ (\omega, v) = \sum_{m=1}^{M} U\ (\omega, r_m)\ J_1\left(\frac{\omega}{v}r_m\right) r_m \Delta r_m \tag{14b}$$

where

$$\Delta\, r_m = \frac{1}{2}\begin{cases} r_2 - r_1, & \text{for m} = 1 \\ r_M - r_{M-1}, & \text{for } m = M \\ r_{m+1} - r_{m-1}, & \text{otherwise} \end{cases}$$

with r_m being pre-defined by the field configuration and M the number of receivers. It is noted that the number of receivers M here is different and is much smaller than the number of wavenumber points M used for the wavefield computation in eq. (11). The Bessel function can be expressed by the sum of two kinds of Hankel functions as

$$J_i\ (x) = \frac{1}{2}\left[H_i^{(1)}(x) + H_i^{(2)}(x)\right] \tag{15}$$

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where $H_i^{(1)}(x)$ and $H_i^{(2)}(x)$ are the first- and second-kind Hankel functions of ith order. Based on Forbriger (2003), Xi *et al.* (2021) pointed out that $H_i^{(1)}(kr)$ and $H_i^{(2)}(kr)$ represent the reverse $(-k)$ and forward $(+k)$ directional velocity scans, respectively. For source to receiver distance in this order, $r_1 < r_2 < \ldots r_M$, we keep only the $H_i^{(1)}(x)$ and eq. (14) can be reduced to

$$\hat{W}_{\text{obs}}\ (\omega, v) = \frac{1}{2} \sum_{m=1}^{M} W(\omega, r_m)\, H_0^{(1)}\left(\frac{\omega}{v} r_m\right) r_m \Delta r_m \tag{16a}$$

$$\hat{U}_{\text{obs}}\ (\omega, v) = \frac{1}{2} \sum_{m=1}^{M} U(\omega, r_m)\, H_1^{(1)}\left(\frac{\omega}{v} r_m\right) r_m \Delta r_m. \tag{16b}$$

It is noted that the action of taking the phase angle of $H_i^{(1)}(\frac{\omega}{v} r_m)$, instead of $e^{i\frac{\omega}{v} r_m}$, and applying the phase-shift method is equivalent to the cylindrical beamformer transform (Zywicki & Rix 2005). Equations (12) and (16) represent two different types of methods for estimating the Green's function and dispersion curves. Their performance in comparison with the theoretical Green's function and dispersion curves is examined below.

3 RESULTS AND DISCUSSION

Green's function (displacement) in the transformed domain for a given layered model can be efficiently computed, but not directly observable. It is independent of the survey geometry and wavefield transformation algorithm used. In order to connect the Green's function to the dispersion curves and observable field quantities, in this section, we first examine the characteristics of theoretical Green's function in different types of earth models. We then connect them to the dispersion curves (including real part of complex roots). Finally, we establish the relation between the Green's function and observed dispersion spectrum (i.e. FVS).

3.1 Homogeneous half-space

3.1.1 Theoretical Green's function versus dispersion curve and leaky surface wave

We first examine the amplitude peaks of the Green's function (displacement) in relation to the dispersion curves in a homogeneous half-space with $V_s = 200$ m s^{-1}. The Green's function does not change with frequency in an isotropic and homogeneous half-space. Plotted Fig. 1 is the amplitude of the Green's function (radial and vertical displacements, i.e. eq. 6) on the surface as a function of the phase velocity. It shows that, besides the well-known main peak corresponding to the Rayleigh pole, there is always a secondary peak at a higher phase velocity, particularly when the Poisson's ratio is high. The numerator and denominator of eq. (6) corresponding to Fig. 1 are shown in Fig. 2. While the modal solution concerns only the denominator, the response solution also depends on the numerator. The secondary peak in the case of low Poisson's ratio (Fig. 1a) is not as significant as Fig. 1(b) due to nearly zero numerator (Fig. 2a) at the phase velocity near the local minimum of the denominator (Fig. 2b). The secondary peak in the Green's function is attributed to the complex conjugate roots in the rationalized Rayleigh equation eq. (9) when the Poisson's ratio is high (Schröder & Scott 2001; Gao *et al.* 2014). Note that there is no complex root in eq. (7) or (8), regardless of the Poisson's ratio of the half-space; the complex conjugates roots can only be found in eq. (9) when Poisson's ratio is greater than 0.263. Furthermore, only one real root corresponding to the main peak can be found in eq. (7) (det[$\boldsymbol{K}$] = 0) and the result is also identical to the root in the analytical Rayleigh equation eq. (8).

Our calculation indicates that the secondary peak in Fig. 1 corresponds to a local minimum in |det[$\boldsymbol{K}$]|, as shown in Figs 2(b) and (d). We emphasize that this local minimum does not reach zero value, that is det[$\boldsymbol{K}$]≠0, even in the purely elastic half-space. In other words, a singular point does not necessarily exist near the local peak of the Green's function! While singular points represent the ideal resonant phenomenon, local peaks are associated with the local minimum of |det[$\boldsymbol{K}$]|. Should eq. (9) be used, extra non-physical roots that do not satisfy the original dispersion equation (eq. 8) can be found as indicated by the vertical dotted line in Fig. 1. These extraneous roots are real for Poisson's ratio (ν) less than 0.26 and become complex conjugates when ν is greater than 0.26. It is interesting that the real part of the complex root at the high Poisson's ratio (0.49) almost aligns with the peak of the vertical component of the Green's function (Fig. 1b). It was hence considered as the phase velocity of the leaky wave in previous studies (Schröder & Scott 2001; Gao *et al.* 2014) using the rationalized Rayleigh equation, eq. (9), and this higher mode (leaky mode) deserves further examination as follows.

The waveforms of the two homogeneous half-spaces in Fig. 1 are computed by eq. (10) and shown in Fig. 3. The Green's function is distributed over the phase velocity (or wavenumber) and is non-zero beyond the singular points (roots of the secular equation). Therefore, the dynamic response wavelet becomes more dispersed as the offset increases for both the fundamental mode and higher mode, although the dispersion curve is non-dispersive (frequency independent). The effect of the second spectral peak leads to a higher velocity wave train, which is more prominent in Figs 3(b) and (d). This is consistent with our observation in Fig. 1 where the radial component has a stronger higher mode energy relative to the fundamental mode than the vertical component. The velocity of the higher-velocity mode observed in Fig. 3(b) is about V_p. With increasing Poisson's ratio, this higher-mode velocity becomes smaller than V_p and close to 2 times V_s when the Poisson's ratio is 0.49, as shown in Fig. 3(d). This higher-velocity wave train, particularly in higher Poisson's ratio case, was referred to as the non-geometric wave in Roth & Holliger (2000) or the leaky surface wave in Gao *et al.* (2014), which was attributed to the complex conjugate roots in the dispersion equation (eq. 9). It is noteworthy that the non-geometric wave can also be clearly observed in the radial component for Poisson's ratio as low as 0.2, as shown in Fig. 3(b). However, these non-geometric waves were mostly observed in high Poisson's ratio cases because typical seismic surveys utilize only vertical geophones.

To closely examine the transition from low Poisson's ratio to high Poisson's ratio, Fig. 4 plots the Green's function for four different Poisson's ratios. Except for the vertical component of Poisson's ratio near 0.5, the real parts of the complex roots of eq. (9) do not really correspond to the higher-mode spectral peaks as previously thought. Extraneous complex conjugate roots can be found in eq. (9) for Poisson's ratio greater than 0.26. Their real parts fall in a small range from 1.89 V_s to 1.96 V_s when Poisson's ratio increases from 0.27 to 0.49. However, as shown in Fig. 4, the energy and apparent velocity of the higher mode significantly increase with increasing Poisson's ratio. Nevertheless, this apparent velocity increase with Poisson's ratio is not as fast as the increase of V_p with increasing Poisson's ratio under constant V_s. The implication of this phenomenon on the *P*-wave first arrival in seismic refraction test is further examined in Fig. 5, in which the waveforms at 50 m offset are compared for different Poisson's ratios. The theoretical first arrivals of *P* waves

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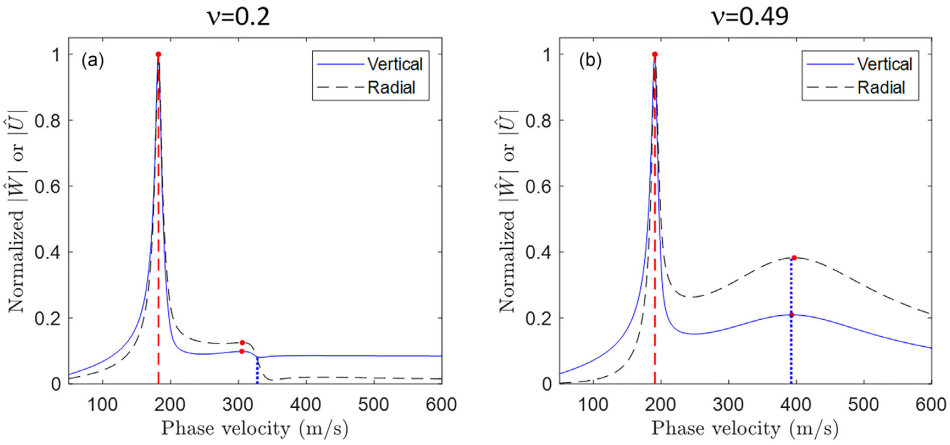


Figure 1. The theoretical frequency-independent Green's function (vertical $\hat{W}(-, v)$ and radial $\hat{U}(-, v)$) of a homogeneous half-space with $V_s = 200$ m s^{-1} for (a) Poisson's ratio $\nu = 0.2$ and (b) Poisson's ratio $\nu = 0.49$. The red dots denote the spectral peaks, the red-dash vertical lines mark the root of eq. (7) (or eq. 8) and the blue dotted lines mark the extraneous root from eq. (9) (we took only the real part when the root is complex for the high Poisson's ratio case).

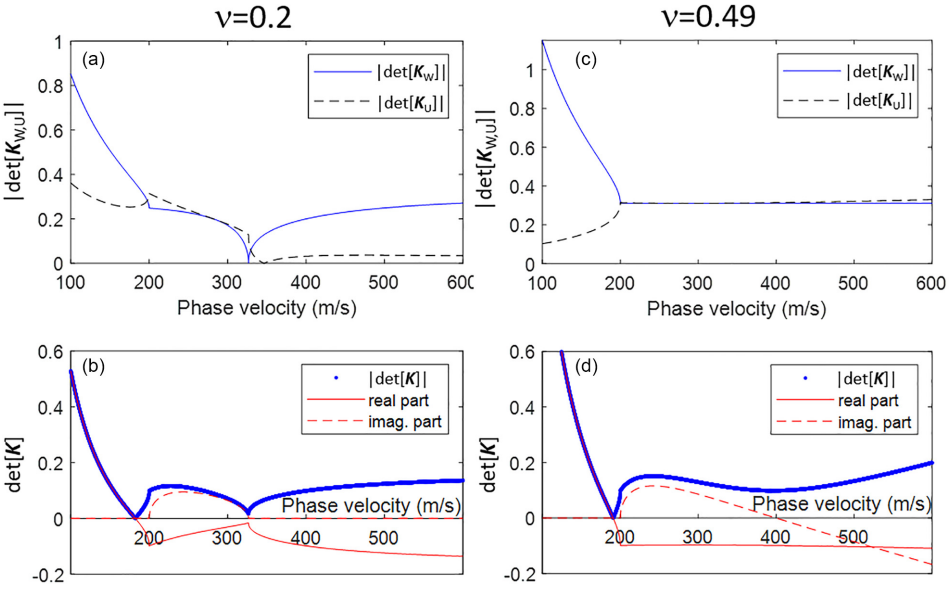


Figure 2. Variations of the numerator and denominator in the Green's functions versus phase velocity. Panels (a) and (b) correspond to Fig. 1(a), and panels (c) and (d) correspond to Fig. 1(b).

are marked by crosses in the horizontal axis. The *P*-wave arrival time relative to the leaky-mode wave train depends on the Poisson's ratio. As a result, it can cause significant ambiguity (uncertainty) in the determination of the first breaks of *P* waves.

3.1.2 Theoretical Green's function versus experimental FVS

In seismic testing, the wavefield is inevitably 'truncated'. As a result, it is not easy to precisely recover the Green's function. In this section, we examine the performance of FVS to approximate the Green's function and the effect of spatial truncation. Special attention will be paid on the FVS peaks in correspondence to the theoretical dispersion curves. The radial component is used since its higher mode is more pronounced. Fig. 6 shows the Green's function amplitude ($|\hat{U}(\omega, v)|$) and the corresponding FVS calculated by the phase-shift method and Fourier–Bessel transform method using 48 receivers with 2 m near offset and 1 m receiver spacing. Both the fundamental mode and the higher mode are captured by both wavefield transformation methods. The truncation effect causes additional spectral oscillations across all frequencies with severe distortion at low frequencies. Therefore, long receiver spread length relative to the wavelength is required in order to minimize the truncation distortion. While the half-space is theoretically non-dispersive, low-frequency dispersion is apparent in the experimental FVS, causing over estimation of low-frequency phase velocity. This is often referred to as the near-field effect. We tested the cylindrical beamforming method on mitigating this effect, the result was similar to the phase-shift method, implying that the low-frequency distortion is mainly attributed to the higher mode and truncation effect. For the same phase velocity, wavelength increases dramatically as frequency decreases below 10 Hz. As a result, the same geophone spread cannot correctly resolve different modes. Furthermore, the Fourier–Bessel transform shows better resolution in mode identification at high frequencies; however, its truncation effect at low frequencies is more severe.

Regardless of the wavefield transformation method utilized, the truncation effect which is location-dependent, is the major source

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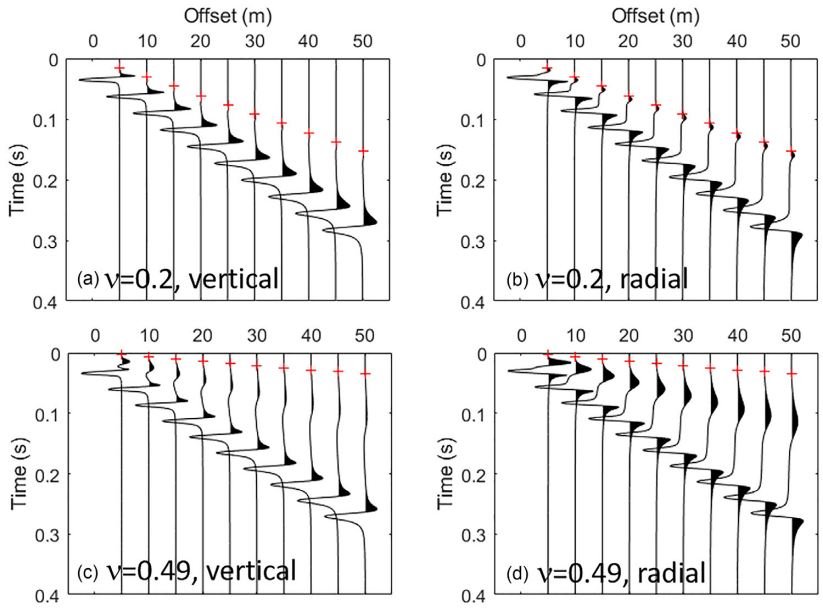


Figure 3. Synthetic waveforms in a homogeneous half-space with $V_s = 200$ m s^{-1}: panels (a) and (b) show, respectively, the vertical and radical components for Poisson's ratio $\nu = 0.2$; (c) and (d) show, respectively, the vertical and radical components for Poisson's ratio $\nu = 0.49$. *P*-wave arrival times are marked as red crosses for comparison.

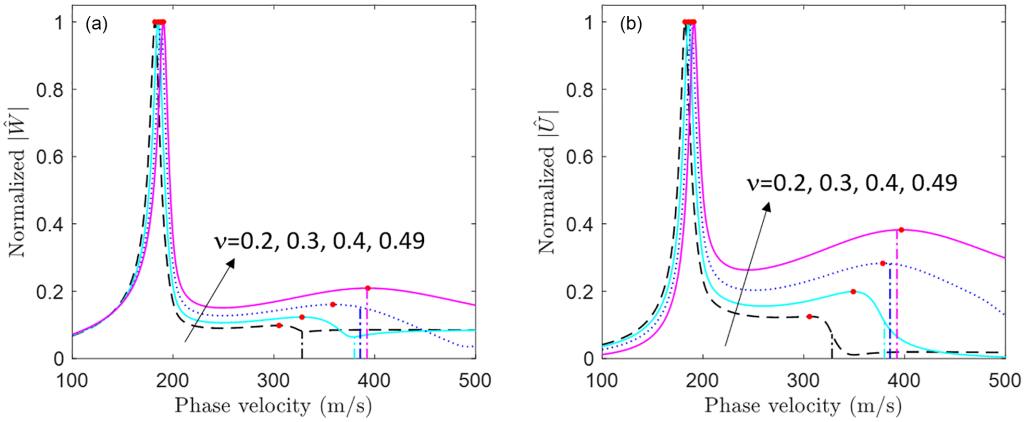


Figure 4. The Green's function of (a) vertical component and (b) radial component for Poisson's ratio 0.2, 0.3, 0.4 and 0.49. The red dots denote the spectral peaks, while the vertical dash-dotted lines mark the extraneous roots (real parts) found in eq. (9). Same colours are used for the same Poisson's ratio.

of uncertainty for experimental estimation of dispersion curves. The spectral curves of Green's function and FVS at 10 and 50 Hz with near offset of 2 and 50 m are shown in Fig. 7 for comparison. It is observed from Fig. 7(b) that the Fourier–Bessel transform method better approximates the Green's function at higher frequencies (compared with Fig. 7a at lower frequencies) as it is more mathematically compatible; however, the deviation from the Green's function is still very significant due to the truncation effect. Figs 7(c) and (d) are the results when the near offset is moved far away at 50 m. The higher mode of the Green's function cannot be observed in the far field when the offsets within 50 m are not included in the wavefield transformation. As also observed in Fig. 3, the higher mode manifests in the near field and fades away with increasing offset. Different modes may manifest in different offset range. While a long receiver array is desirable, especially for low-frequency (long-wavelength) data, it is hard to say what the optimal near offset would be. Thus, to reduce the uncertainty from the truncation effect, the receiver layout should be considered in the inversion, instead of relying on ensuring a minimum near offset (Xu *et al.* 2006).

With the efficiently calculated Green's function and complete dynamic response, surface seismic testing can include all waves observed on the ground surface. We don't have to emphasize the 'surface wave' (Rayleigh and Love) alone. It is often considered that surface waves are not 'developed' at the vicinity of source point because they are formed through interference of (*P* and *SV*) body waves generated from reflections and refractions, and thus their occurrence requires a certain minimum distance from the source. Such a statement is not precise. For instance, Rayleigh wave exists in a homogeneous half-space, in which there is no reflection and no refraction. Surface waves exit at all offsets while the near field effect is more of a 3D effect. As near offset increases, the wavefield can be approximated by plane waves. The deviation of dispersion energy from theoretical dispersion curve shown in Fig. 6 is more due to the effect of leaky mode and spatial truncation, rather than the 3-D effect. Another type of leaky mode, the fast *P–SV* guided

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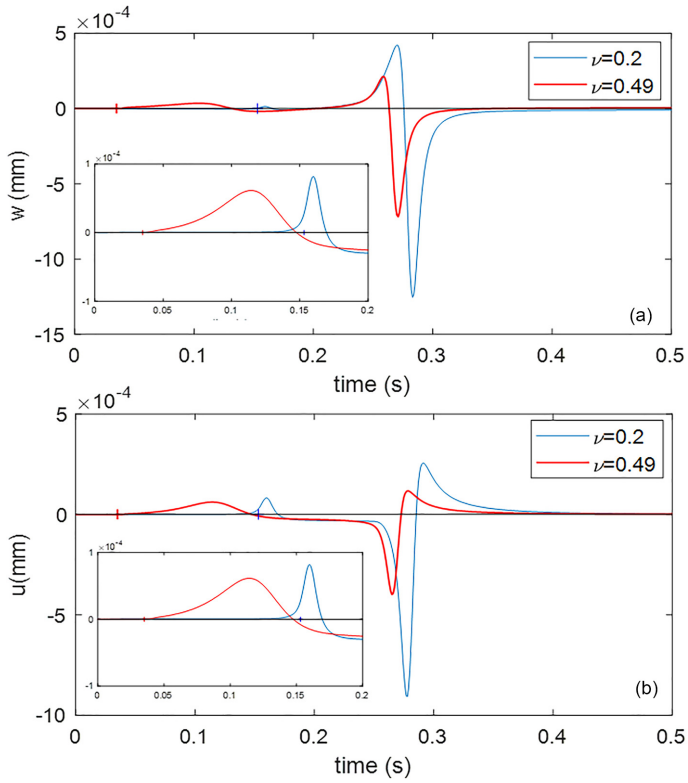


Figure 5. (a) Vertical and (b) radial components of the waveforms at 50 offset for two Poisson's ratio ($\nu = 0.2$ and 0.49). The theoretical *P*-wave arrival times are marked by cross in the horizontal axis.

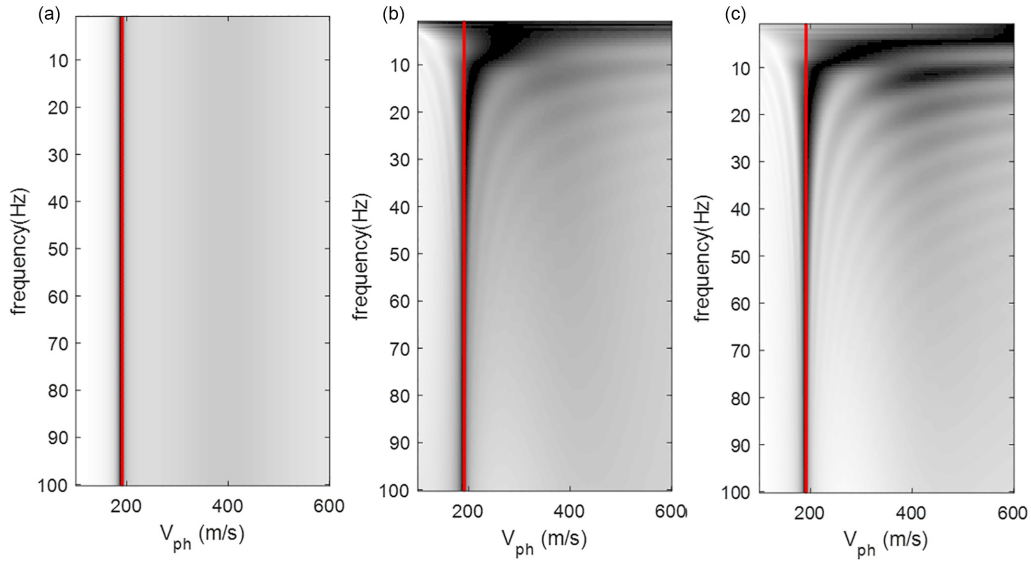


Figure 6 (a) The theoretical Green's function amplitude for the radial component and the corresponding FVS estimated by (b) the phase-shift method and (c) the Fourier–Bessel transform method, from 48 receivers with 2 m near offset and 1 m receiver spacing in a half-space with $V_s = 200$ m s^{-1} and $\nu = 0.49$.

waves (García-Jerez & Sánchez-Sesma 2015), will be revealed in the layered elastic medium as follows.

3.2 Two-layer models

3.2.1 Dispersion curve gap in inverse Vs profile

In layered structures, surface geophones can receive reflected waves and refracted head waves in addition to the surface waves. Excluding the effect of spatial truncation from limited spatial sampling, we first examine the theoretical Green's function and dispersion curves. Fig. 8 shows the amplitude spectrums of the theoretical Green's function for two-layer models (with first layer thickness of 5 m) as well as that of the homogeneous half-space for comparison (where Fig. 8c is discussed in the last subsection). As the higher mode is more prominent in the radial component, only the radial component is shown in Fig. 8. The model is selected such that the V_P equals 1500 m s^{-1} in all the layers and V_S of the

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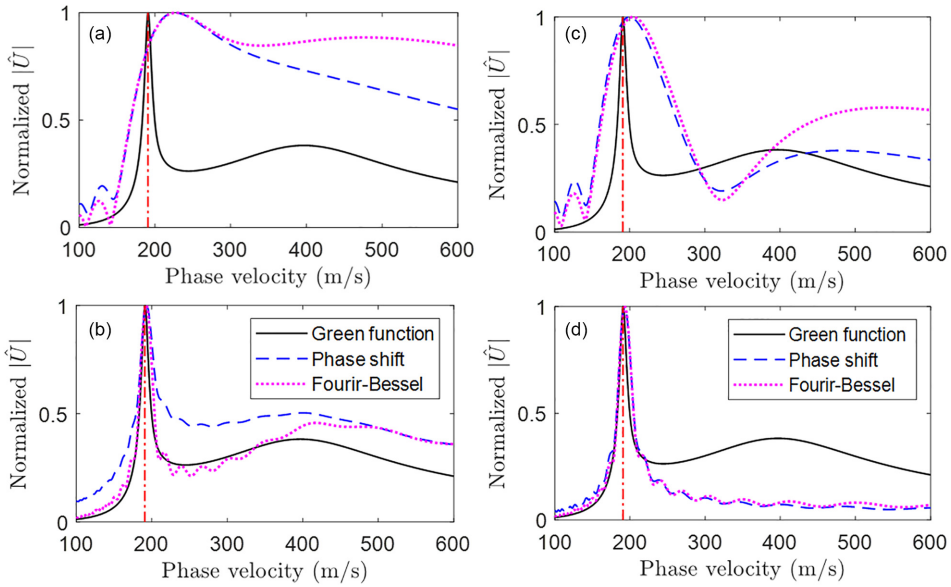


Figure 7 Panles (a) and (b) show the spectral curves of Green's function and FVS at 10 and 50 Hz for a near offset of 2 m; panels (c) and (d) show the same plots for a near offset of 50 m. The FVS are estimated from 48 receivers at 1 m spacing in a half-space with $V_s = 200$ m s^{-1} and $\nu = 0.49$.

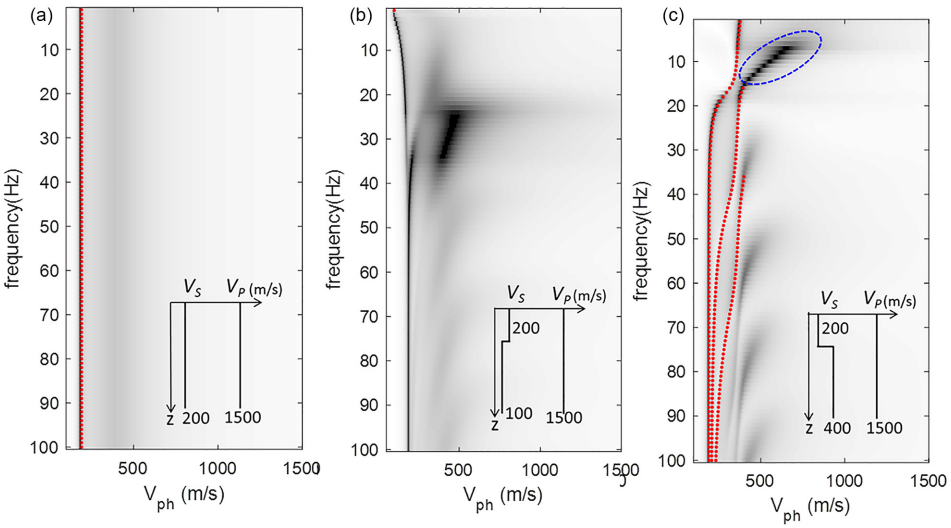


Figure 8 The radial-component amplitude spectrum of the theoretical Green's function: (a) homogeneous half-space with $V_S = 200$ m s^{-1} and $V_P = 1500$ m s^{-1}; (b) two-layer structure with 5 m layer thickness, V_S= [200 100] m s^{-1}, and homogeneous $V_P = 1500$ m s^{-1}; (c) two-layer structure with 5 m layer thickness, V_S= [200 400] m s^{-1}, and homogeneous $V_P = 1500$ m s^{-1}. The red dots are the dispersion curves of the first three modes.

surface layer is 200 m s^{-1} in all the three models. Thus, there are no pure *P* wave refractions and reflections involved. The result of the homogeneous half-space in Fig. 8(a) is basically the same as Fig. 6(a) but covers the phase velocity up to V_P for comparison with the two-layer models. Fig. 8(b) shows the result for the case of a lower-V_S half-space, in which there are no refracted head waves from *P* waves and *S* waves since V_P is homogeneous and the *S*-wave velocity of the bottom layer is lower than the top layer. For this inverse V_S profile (i.e. higher-V_S layer over lower-V_S half-space), there is a threshold frequency below which the fundamental mode dominates. If the surface layer is not the softest layer, higher modes dominate at high frequencies (shorter wavelengths). Inferring from various simulations, the transition from the fundamental mode to higher mode occurs at the wavelength approximately equal to the depth to the lower homogeneous half-space (Lin *et al.* 2021). Except for the difference in Rayleigh wave dispersion, differences between Figs 8(b) and (a) (the homogeneous half-space) arise from shear wave reflections and possibly the converted *P* waves at the layer interface. As a result, the uniform dispersion energy of the surface leaky mode shown in Fig. 8(a) becomes disruptive and amplified at some frequencies.

It is noteworthy that the apparent Rayleigh modes observed in Fig. 8(b) are not necessarily the free vibration modes (i.e. det[$\boldsymbol{K}$] = 0). When the V_S of the underlying half-space is smaller than those of the upper layers (typical situation for a pavement profile) there is a frequency range between which no real wavenumbers

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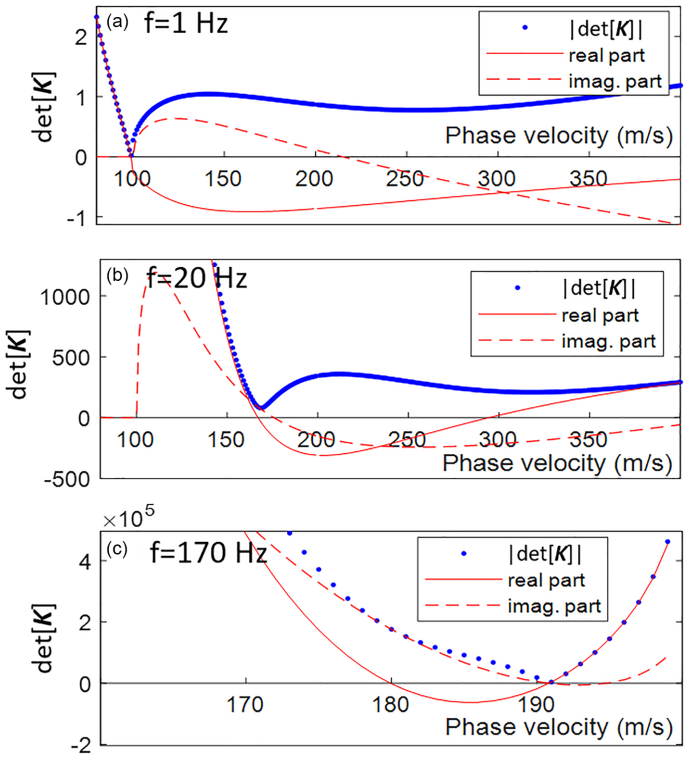


Figure 9. Variation of the absolute value, and real and complex parts of det [***K***] versus phase velocity of the earth model in Fig. 8(b) for three frequencies: (a) $f = 1$ Hz, (b) $f = 20$ Hz and (c) $f = 170$ Hz.

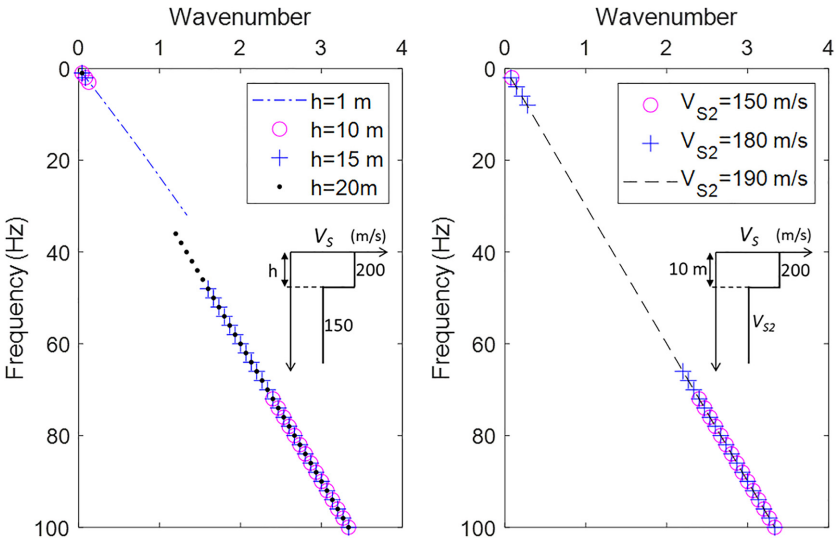


Figure 10 Dispersion curve gap in a two-layer model: (a) different gap as a function of the top layer thickness with constant V_S profile ($V_{S1} = 200$ m s^{-1} and $V_{S2} = 150$ m s^{-1}); (b) different gap as a function of the V_S in the lower half-space with constant layer thickness (10 m) and top layer V_S (200 m s^{-1}). The Poisson's ratio is 0.4 for all cases.

exist to make the determinant zero. In the particular case shown in Fig. 8(b), this frequency range is between 1 and 170 Hz. Fig. 9 shows the corresponding det[***K***] versus phase velocity for frequency at 1, 20 and 170 Hz. Below 1 Hz, the imaginary part of det[***K***] is zero and the root of eq. (7) is the same as the zero crossing of the real part of det[***K***]. For frequency above 1 Hz but below 170 Hz, the real and imaginary parts of det[***K***] never reach zero at the same phase velocity. In this frequency range, the dispersion energy shown in Fig. 8(b) corresponds to the local minimum of det[***K***], as shown in Fig. 9(b). For frequency above 170 Hz, roots of eq. (7) appears again as the real and imaginary parts of det[***K***] have the same zero crossing.

This special wave feature we just observed and analysed within the given frequency range is the so-called elastic wave bandgap within it the elastic wave cannot pass. Such a feature is related to the combination of material properties, thickness as well the frequency

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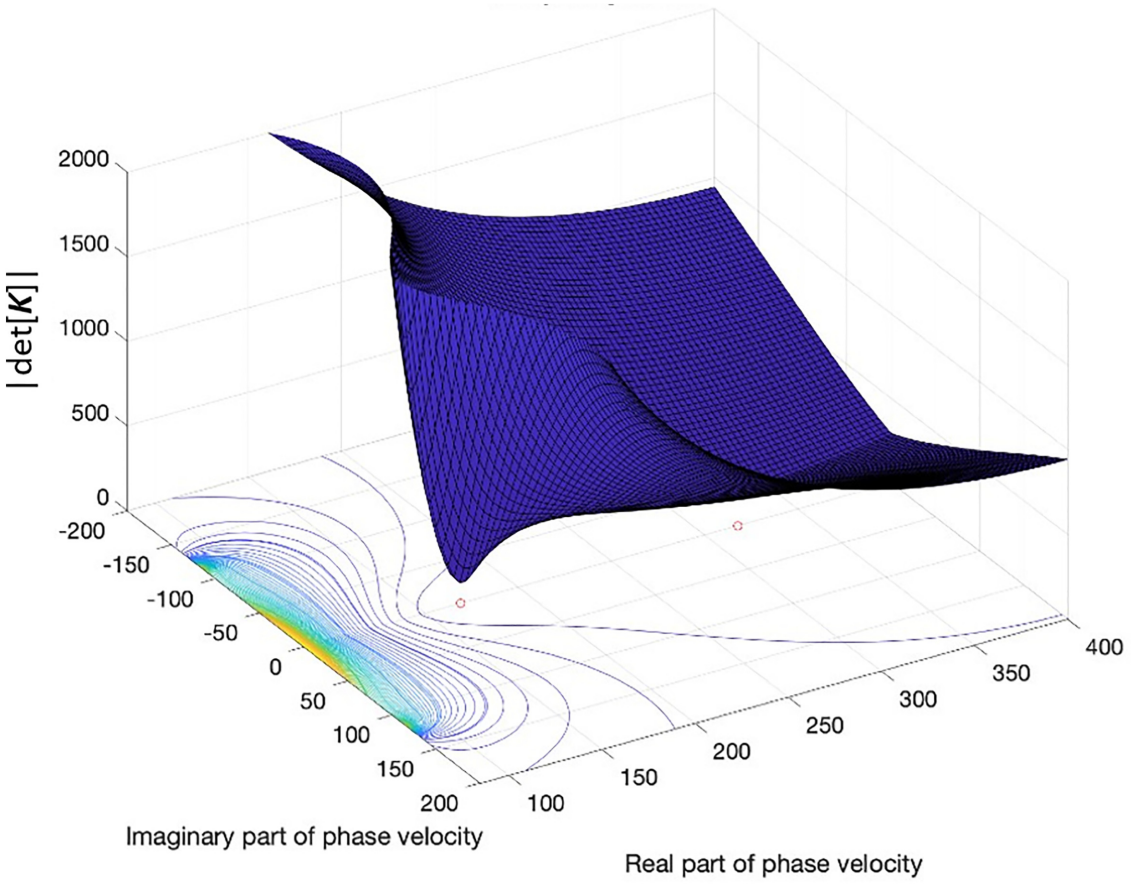


Figure 11. The |det[***K***]| versus complex phase velocity of the earth model in Fig. 8(b) for frequency $f = 20$ Hz, which corresponds to Fig. 9(b).

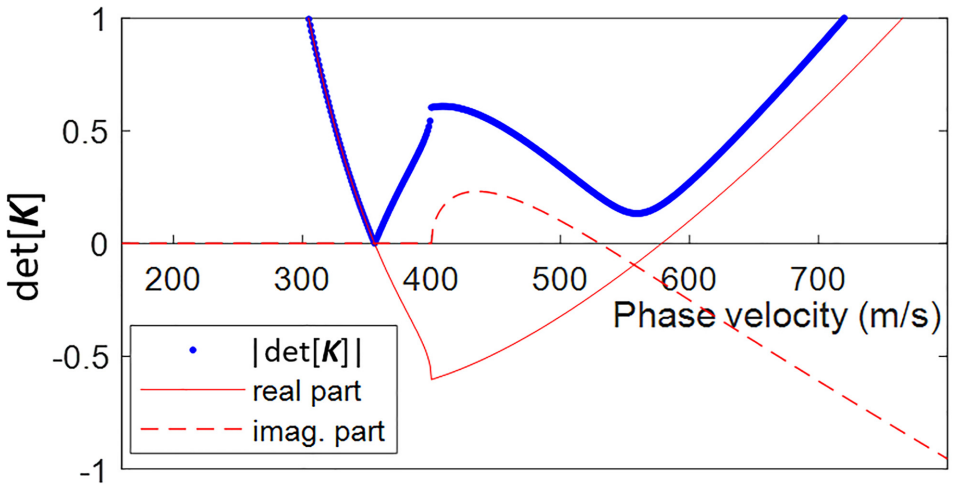


Figure 12. The det[***K***] versus phase velocity of the earth model in Fig. 8(c) at frequency $f = 10$ Hz.

range under consideration, and could have many applications in civil engineering, particular in earthquake engineering where one would like to design a special layered metamaterial to counteract the earthquake induced adverse wave (i.e. Chen *et al.* 2019; Li *et al.* 2020; Mu *et al.* 2020). Fig. 10 illustrates the dispersion curve gap in the *f*–*k* domain as a function of the layer thickness and velocity contrast in the case of a two-layer model. As top layer thickness increases, the bandgap is reduced in a way that both the lower and upper bounds of the gap decreases. As velocity contrast decreases, the bandgap is also reduced but in a way that the lower bound increases and the upper bound decreases. When the velocity contrast is less than 10 per cent, there is no bandgap and the dispersion curve becomes continuous.

In the frequency gap where there are no real roots for eq. (7), a practical solution is to select the real values of the phase velocity (or wavenumber) that correspond to the zeros of real (det[***K***]). As shown

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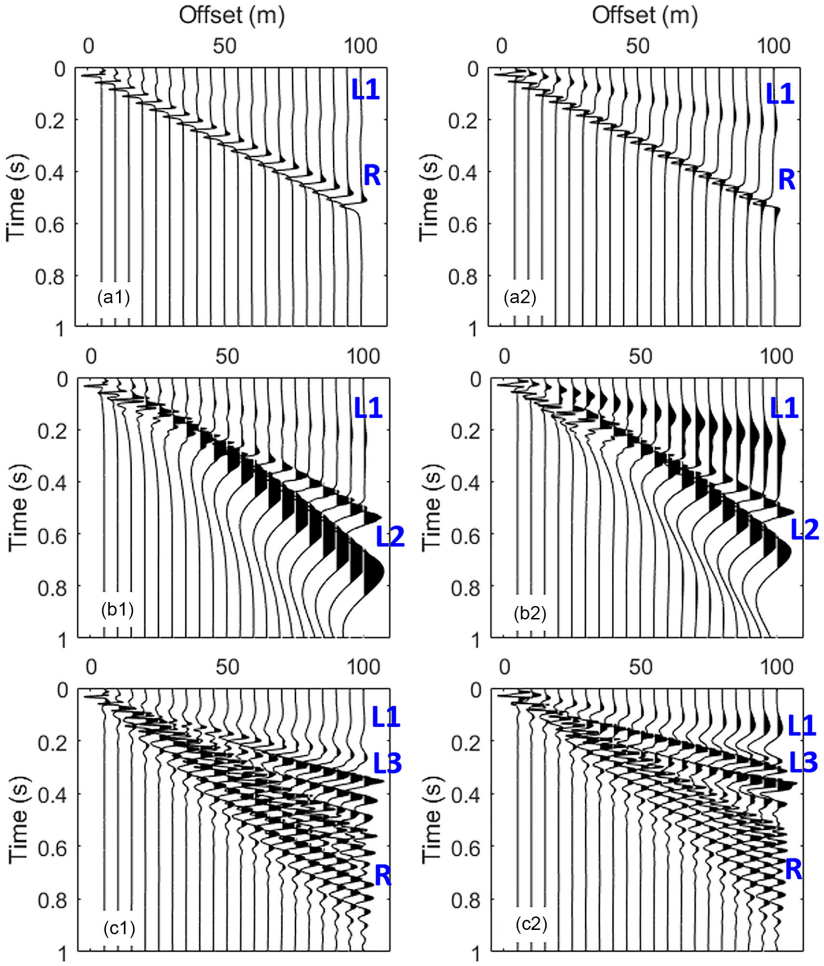


Figure 13. Time-domain responses of the three cases in Fig. 8. Panels (a)–(c) here correspond to Figs 8(a)–(c), and numbers 1 and 2 indicate the vertical and radial components, respectively. Letters 'R', 'L1', 'L2' and 'L3' represent for Rayleigh wave, surface leaky wave, apparent Rayleigh wave and fast-guided *P–SV* wave.

in Fig. 9, this approach is a good approximation, but may result in extraneous roots as shown in Fig. 9(c). Alternatively, it was suggested to determine the complex roots of the characteristic equation and then only use the real part as the Rayleigh-wave phase velocity (Pan *et al.* 2013). Fig. 9(b) shows $|\det[\boldsymbol{K}]|$ as a function of the phase velocity for the case without a real root and Fig. 11 further shows that there is actually no complex root either. When eq. (7) is used in modal analysis, there are no complex roots in purely elastic isotropic media. Complex roots arise when the medium becomes viscoelastic. While in this study, leaking modes are separately represented by local minima of $\det[\boldsymbol{K}]$, complex roots in other existing modal models may represent either free mode of viscoelastic medium or leaking mode of purely elastic medium.

3.2.2 Fast P–SV guided wave in layered medium with higher-V_S half-space

In the case of normal V_S profile (i.e. lower-V_S layer over higher-V_S half-space), the Rayleigh wave dispersion shows higher velocity with decreasing frequency as expected, and real roots of higher modes can be found up to the maximum V_S. Furthermore, with the higher-V_S half-space, the *P–SV* body waves could be guided into the lower-V_S layer, and at certain frequency, can propagate with a phase velocity greater than the maximum *S*-wave velocity. This is demonstrated by the higher-velocity dispersion at the lower-end frequencies as indicated by the dashed ellipse in Fig. 8(c). This part of higher-velocity dispersion peaks corresponds to the non-geometric $\bar{\text{P}}S$ wave (Roth & Holliger 2000; Sun *et al.* 2021). It was attributed to the leaky wave arisen from high Poisson's ratio by Gao *et al.* (2014), yet the three cases in Fig. 8 all have the same surface layer with high Poisson's ratio of 0.49. Furthermore, as will be shown later, this feature also exists in the case of low Poisson's ratio. Therefore, what we observed in Fig. 8(c) where the Rayleigh-like dispersion spreads significantly faster than the leaky mode in the low-frequency range, is not associated with the high Poisson's ratio, but rather is attributed by the high V_S contrast between the top layer and the lower half-space (García-Jerez & Sánchez-Sesma 2015). García-Jerez & Sánchez-Sesma (2015),, further explained this feature in terms of contour integral as a result of the relative contributions from Rayleigh pole and branch cut. Here we illustrate this feature simply in terms of the Green's function. Again, from Fig. 8(c), the fast-guided wave is observed at frequency $f = 10$ Hz. At this frequency, we now examine the function $\det[\boldsymbol{K}]$ versus phase velocity as in Fig. 12. Clearly, it shows

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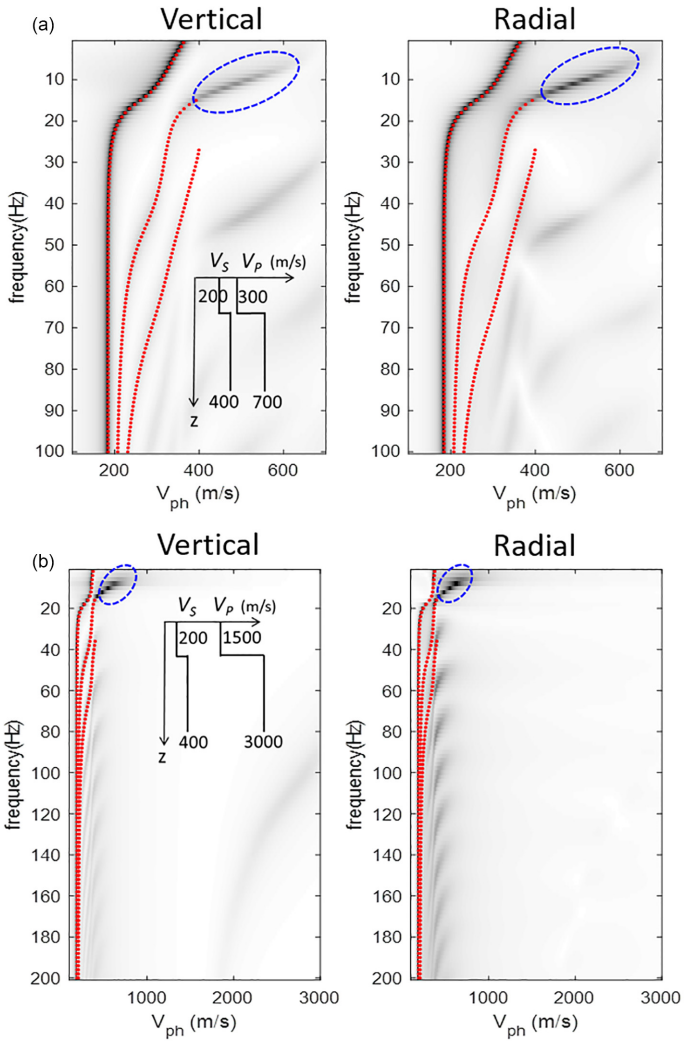


Figure 14 Amplitude spectra of Green's function of two-layer models with the same layer thickness (5 m) and *S*-wave velocity contrast (200 m s^{-1} over 400 m s^{-1}) but with different Poisson's ratios (a) 0.26 (350 m s^{-1} over 700 m s^{-1} for V_P) and (b) 0.49 (1500 m s^{-1} over 3000 m s^{-1} for V_P).

that the fundamental (Rayleigh) mode velocity (i.e. the root of the determinant) is at 356 m s^{-1} and a local minimum at 560 m s^{-1}. The latter is the velocity of the fast-guided *P–SV* wave which corresponds to the major dispersion energy in Fig. 8(c). In this case, the spectral amplitude of the fundamental Rayleigh mode is lower than the fast-guided wave associated with the local minimum of det[***K***]. This is because the numerator in eq. (6) is also nearly zero at the root of eq. (7).

Up to this point, we have seen three different cases where significant response of the Green's function (peak in dispersion image) is not located near the root of the dispersion equation (i.e. not the Rayleigh pole). They can all be referred to as a type of leaky mode. However, we distinguish them as (1) the surface leaky wave that exists even in homogeneous half-space (more pronounced in high Poisson's ratio and in radial component), **(2)** the apparent Rayleigh mode within a frequency gap in a layered medium with a low-velocity half-space; and **(3)** the fast-guided *P–SV* wave in the layered medium with a high V_S contrast between the upper layers and the lower half-space. These leaky waves are induced by different conditions, but they can all be clearly identified by the local minima of det[***K***]. Therefore, by finding the local minima of det[***K***], one can find all normal modes as well as the leaky modes. To further help understand their differences, the time-domain responses of the three cases in Fig. 8 are shown in Fig. 13. Clearly, the surface leaky waves exit in all the three cases, while the time-domain features of the apparent Rayleigh mode and the fast-guided *P–SV* wave can be uniquely observed in Figs 13(b) and (c), respectively.

3.2.3 *Conditions for mode kissing*

It can be shown that the manifestation of the fast-guided *P–SV* wave is mainly due to the velocity contrast between the top layers and high-velocity half-space. Fig. 14 shows the amplitude spectrum of Green's function for two-layer models with the same V_S profile (V_{S1} = 200 m s^{-1} over V_{S2} = 400 m s^{-1}, first layer thickness 5 m) but different Poisson's ratios. Fig. 14(a) shows the vertical and radial components for Poisson's ratio at 0.26 (with V_{P1}= 350 m s^{-1} over V_{P2}= 700 m s^{-1}), and Fig. 14(b) shows the vertical and radial components for Poisson's ratio at 0.49 (with V_{P1}= 1500 m s^{-1} over V_{P2}= 3000 m s^{-1}). Both results show the spectral energy due to the fast-guided *P–SV* waves as indicated by the dashed ellipse. This demonstrates that the fast-guided *P–SV* waves can also occur

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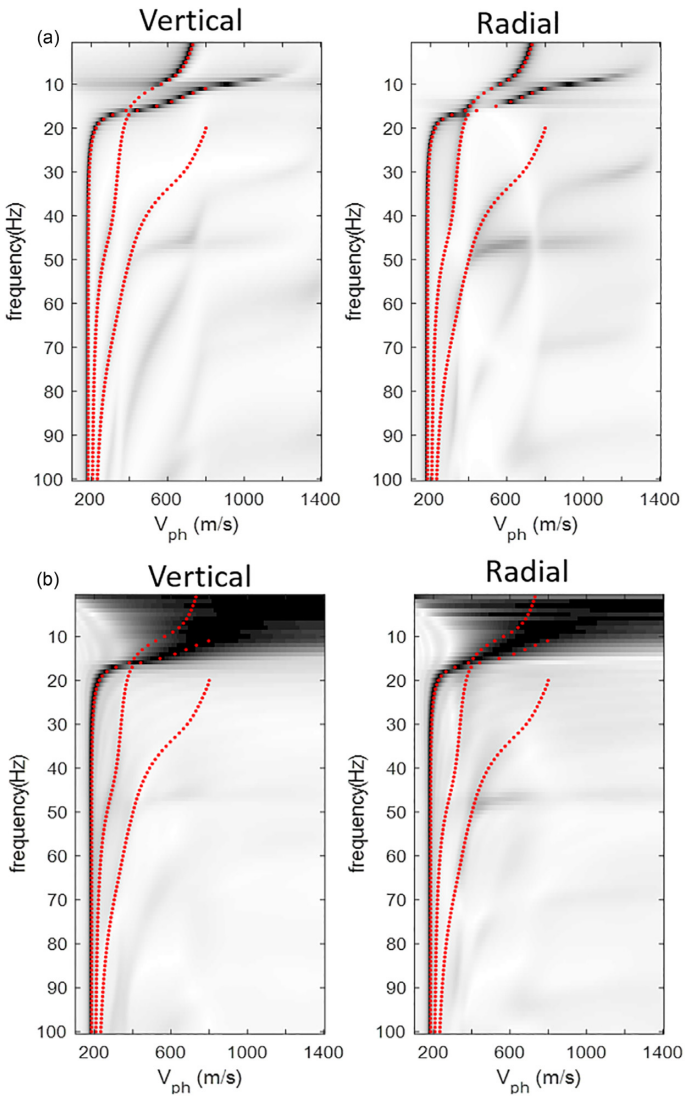


Figure 15. (a) The vertical and radial components of the Green's function amplitude spectrum in a two-layer model with high V_S-velocity contrast (200 m s^{-1} over 800 m s^{-1}) with constant Poisson's ratio 0.26 and first-layer thickness 5 m. (b) The corresponding FVS using 48-channel data with near offset 10 m and receiver spacing 1 m by the phase-shift method.

in the case of low Poisson's ratio, although it is more dominant when the Poisson's ratio is higher (Gao *et al.* 2016). Unlike Fig. 8, the V_P is not homogeneous in Fig. 14. As such, there is additional effect of reflections and refractions from *P* waves. In the case of high Poisson's ratio (V_P structure with 1500 m s^{-1} over 3000 m s^{-1}), V_P is much higher than the velocity of the fast-guided *P–SV* waves. The frequency range is extended to 200 Hz in Fig. 14(b). A separate high-velocity dispersion energy can be observed between 1500 and 3000 m s^{-1} in the vertical component (Fig. 14b Vertical). This is similar to what was observed and called the residual guided wave in the phase velocity region ($V_{P1} < V < V_{P2}$) by Sun *et al.* (2021). For low Poisson's ratio case, *S*-wave velocity range overlaps with *P*-wave velocity range (i.e. $V_{S2} > V_{P1}$). The dispersion energies at higher frequency and phase velocity greater than V_{S2} are either the other branch of fast-guided *P–SV* waves or the residual guided waves. In summary, in addition to the Rayleigh-wave dispersion, the Green's function in the transformed domain as shown in Figs 8 and 14 contains all the wave phenomena, illustrating clearly the accumulated effects of the Rayleigh wave, surface leaky wave, fast-guided *P–SV* (from reflected and refracted *S* waves) and residual guided waves (from reflected and refracted *P* waves).

The dispersion energy of the fast-guided *P–SV* waves appears as an extension of the first higher mode towards lower frequency and higher velocity. The energy of the fast-guided *P–SV* waves increases as the V_S-profile contrast increases, the Poisson's ratio increases, or the layer thickness (above the high-velocity lower half-space) decreases. Experimental FVS has limited spectral resolution, especially at low frequencies where the fast-guided *P–SV* waves are prominent. The spatial truncation due to limited receiver spreading makes it difficult to separate the fundamental mode from the higher mode including the fast-guided *P–SV* waves. Mode kissing in the FVS may occur when velocity contrast is high or when the Poisson's ratio is high. We illustrate these below.

Fig. 15 shows the mode kissing in a two-layer model with high velocity contrast ($V_{S1} = 200$ m s^{-1}, $V_{S2} = 800$ m s^{-1}, with the first layer thickness = 5 m) and a low Poisson's ratio (0.26). As can be observed from the amplitude spectrum of the Green's function in Fig. 15(a), the two modes almost touch each other near frequency

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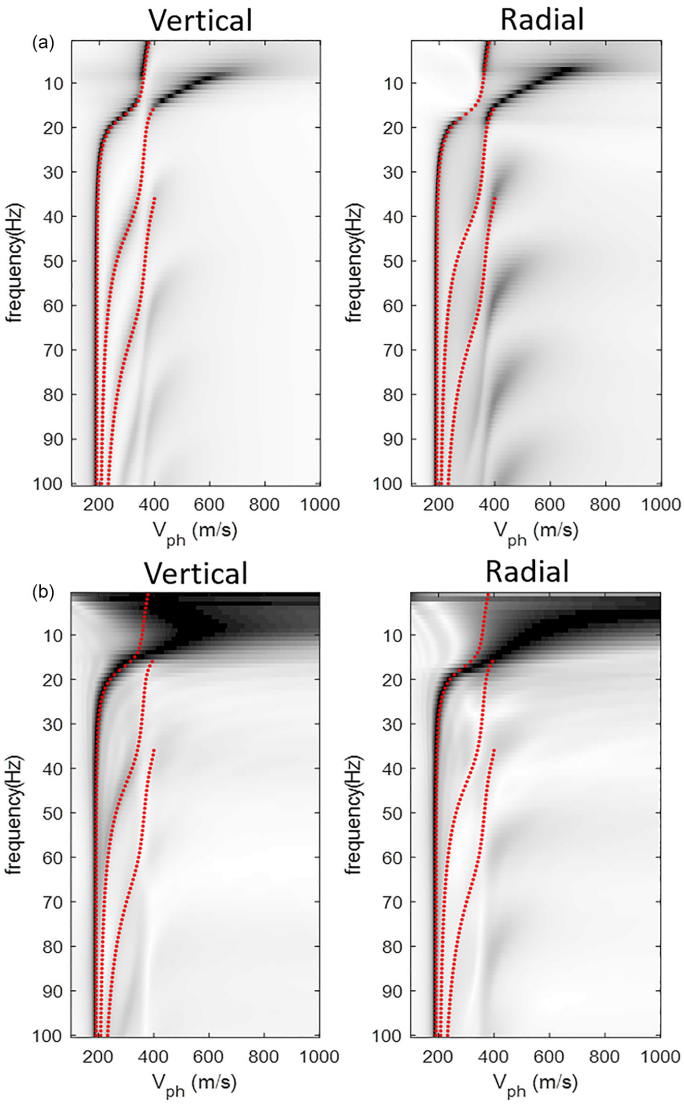


Figure 16. (a) The vertical and radial components of the Green's function amplitude spectrum in a two-layer model of milder V_S-velocity contrast (200 m s^{-1} over 400 m s^{-1}) with constant Poisson's ratio 0.49 and first-layer thickness 5 m. (b) The corresponding FVS using 48-channel data with near offset 10 m and receiver spacing 1 m by the phase-shift method.

at 15 Hz and the first higher mode dominates over the fundamental mode between 10 and 15 Hz, especially for the vertical component in Fig. 15(a, Vertical). We use a typical field layout with near offset 10 m, 48 receivers, and receiver spacing of 1 m, the FVS using the phase-shift method is shown in Fig. 15(b). For frequency below 15 Hz, the experimental FVS is dominated by the first higher mode and it basically connects with the higher-frequency fundamental mode. Without understanding this mode kissing feature, an erroneous fundamental-mode dispersion curve would be obtained from such an experimental result.

The mode kissing could disappear if the velocity contrast is reduced (e.g. reducing V_{S2} in Fig. 15 from 800 m s^{-1} to 400 m s^{-1}). However, as the Poisson's ratio of the surface layer increases, the velocity contrast that can induce mode kissing is reduced. Fig. 16 illustrates the mode kissing in a case where V_{S1} is the same as in Fig. 15, but the Poisson's ratio is increased to 0.49 while V_{S2} is reduced to 400 m s^{-1}. As shown in Fig. 16(a), the two normal modes are clearly separated from each other near $f = 15$ Hz, in contract to those in Fig. 15. However, the energy of the fast-guided *P–SV* waves is stronger in the case of high Poisson's ratio. As a result, the experimental FVS exhibits similar mode kissing. In both cases (Figs 15 and 16), real roots cannot be found for velocity beyond the maximum shear-wave velocity.

Therefore, based on our two-layer models, we find that the fast-guided *P–SV* waves are manifested when there is a strong V_S contrast, exhibiting a Rayleigh-like dispersion mode which can spread significantly faster than the Rayleigh wave and the maximum V_S of the velocity profile. This new leaky mode appears as an extension of the first higher mode towards lower frequency and higher velocity. Mode kissing can occur in the experimental FVS when V_S contrast is high. As the Poisson's ratio of the surface layer increases, the velocity contrast that can induce mode kissing is reduced. When mode kissing does occur, it is difficult to perceive its occurrence or identify the osculation point between the two mode curves without prior information. The ultimate countermeasure of mode kissing in the context of surface wave inversion is to consider all wave phenomena in the forward kernel function, that is the full-field Green's function.

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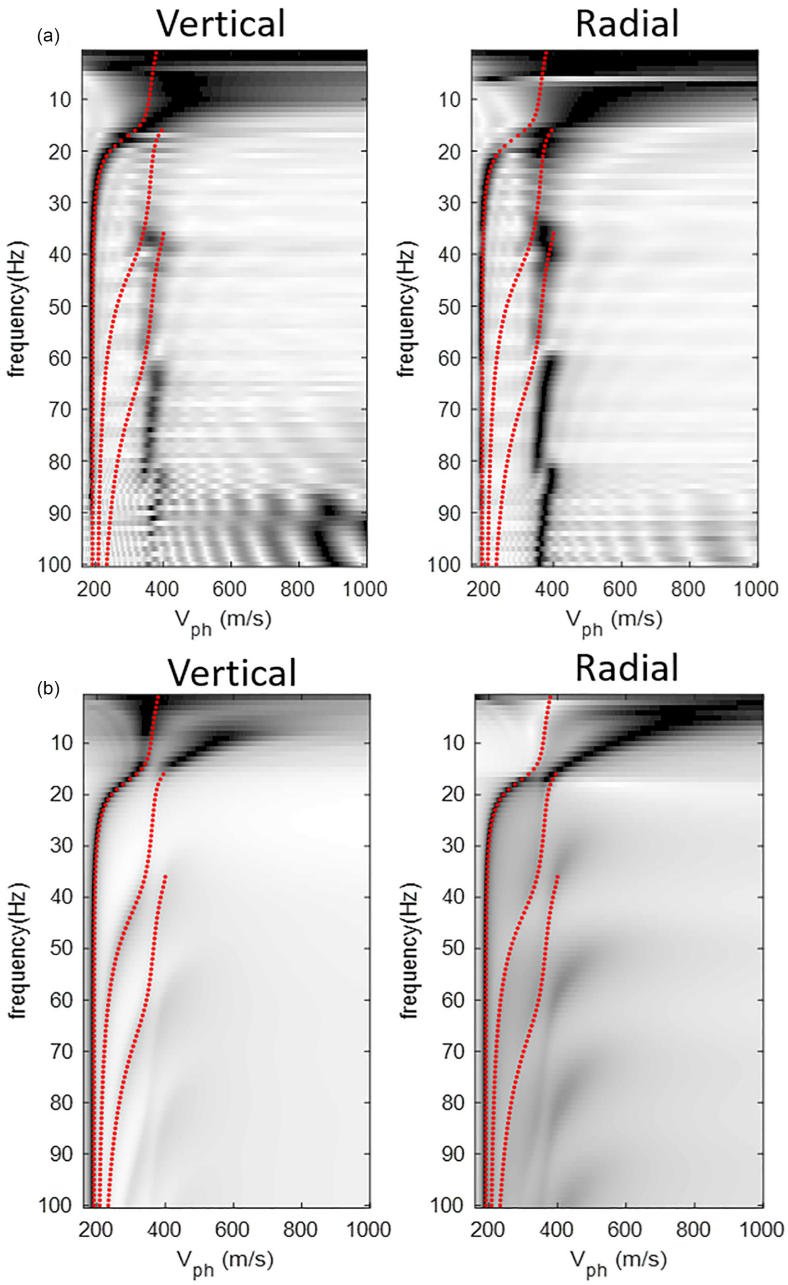


Figure 17. The experimental FVS corresponding to the theoretical Green's function in Fig. 16(a) using the phase-shift method. (a) is obtained by 48-channel receivers with 1 m spacing and near offset 100 m and (b) is obtained by 192 receivers spanning from 1 to 192 m.

3.2.4 Theoretical Green's function versus experimental FVS

We use the earth model of Fig. 16 to re-examine the difference between the theoretical Green's function and the experimental FVS. The experimental FVS shown in Fig. 16(b) is obtained by the phase-shift method using 48-channel receivers with near offset 10 m. Fig. 17(a) shows the results when the near offset of the 48-channel receivers is increased to 100 m. Comparing Figs 17(a) and 16(b), we note that, in the far field with the same receiver array, the higher modes are more prominent. Even with an unreasonably large near offset, Fig. 17(a) shows that the slowly attenuated guided *P–SV* waves cannot be avoided by having a near offset over certain minimum offset. This contradicts the conception that the near field is contaminated by body waves. The result of a more ideal testing layout in which 192 receivers span from 1 to 192 m is shown in Fig. 17(b). The wide spatial coverage of the wavefield does result in much better approximation of the theoretical Green's function. Re-analysing the data in Fig. 17 using the Fourier–Bessel transformation, the results are shown in Fig. 18. In general, the Fourier–Bessel method resolves different modes more clearly especially when a large receiver array is used. When a typical 48-channel array is used, the truncation effect at low frequencies is more significant and different from the phase-shift method as also shown in the homogeneous half-space case. Therefore, in order to improve the surface wave inversion, the receiver locations should be considered, regardless of the type of wavefield transformation method used.

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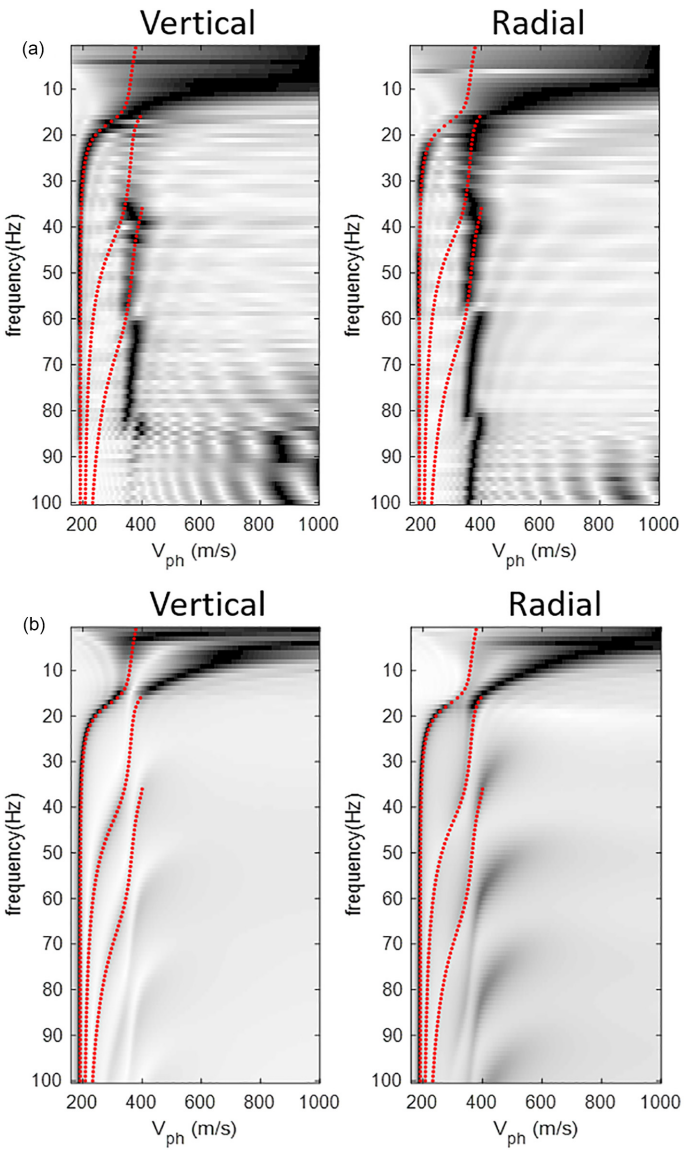


Figure 18. The experimental FVS corresponding to the theoretical Green's function in Fig. 16(a) using the Fourier–Bessel method. (a) is obtained by 48-channel receivers with 1 m spacing and near offset 100 m and (b) is obtained by 192 receivers spanning from 1 to 192 m.

4 CONCLUSIONS

There are a few misconceptions about surface wave dispersions, especially those energy peaks in the dispersion spectrum that do not correspond to roots of dispersion equation. In this study, the dispersion equation, Green's function and dynamic response are all derived in terms of the global stiffness matrix ($\boldsymbol{K}$) to establish the explicit and direct relationship among them. Several new findings from this study are as follows.

(1) It is first clarified that complex roots do not really exist in homogeneous elastic half-space. Extraneous roots are resulted from rationalization of the dispersion equation. These extraneous roots are complex when the Poisson's ratio is high and their real parts do not really correspond to the spectral peak of Green's functions.

(2) Leaky modes (i.e. strong energy peaks in the dispersion image arise at locations where roots of dispersion equation do not exist) can be induced by several different conditions, but they can all be clearly identified by the local minima of det[$\boldsymbol{K}$], while the dispersion equation is det[$\boldsymbol{K}$] = 0. Therefore, by finding the local minima of det[$\boldsymbol{K}$], one can find all normal modes as well as the leaky modes. In this regard, leaky modes can be simply treated as higher modes.

(3) Three distinct types of leaky modes are revealed in this study. We distinguish them as (a) the intrinsic surface leaky wave, (b) the apparent Rayleigh mode within a frequency gap in a layered medium with a low-velocity half-space and (c) the fast-guided *P–SV* wave in the layered medium with a high V_S contrast between the upper layers and the lower half-space.

(i) The surface leaky mode is intrinsic in vertically loaded surface waves. It exists even in homogeneous half-space and is more pronounced in the radial component and when surface Poisson's ratio is high. The surface leaky wave alone does not cause misidentification of modes.

(ii) When the V_S of the underlying half-space is smaller than those of the upper layers (typical situation for a pavement profile), Rayleigh mode can be clearly observed but there are missing roots in

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the dispersion equation (i.e. a bandgap in the dispersion curve). This has been a major issue when MASW is to be applied to pavement testing. The correct countermeasure is to find the apparent Rayleigh mode by local minima of det[***K***] or inversion based on dynamic response.

(iii) When the V_S of the underlying half-space is much higher than those of the upper layers, the fast-guided *P–SV* wave is induced in the low-frequency range and appears as an extension of the first higher mode towards lower frequency and higher velocity. As velocity contrast increases, the first higher mode moves closer to the fundamental mode to nearly have an osculation point. The fast-guided *P–SV* wave has a strong energy (increasing as V_S-profile contrast and Poisson's ratio increase) and is located in the low-frequency range where the receiver truncation effect is significant; hence, the high-frequency fundamental mode may be connected to the low-frequency first higher mode (including the fast-guided *P–SV* waves) in the experimental dispersion image. To avoid misinterpretation of this mode kissing condition and take advantage of its presence, the guided wave can be included in the inversion by considering the actual dynamic responses in the receiver array.

(4) Regardless of the wavefield transformation method utilized, the truncation effect is the major source of uncertainty for experimental estimation of dispersion curves. The truncation effect is both location- and model-dependent, and as such, it is not possible to define a unique optimal near offset. To reduce the uncertainty from the truncation effect, the receiver layout should be considered in the inversion of dynamic response, instead of relying on ensuring a minimum near offset. With the efficiently calculated Green's function and complete dynamic response, all wave phenomena (including different types of leaky waves), receiver locations and wavefield transformation method can be fully taken into account in the inversion.

This study used only two-layer models to obtain and illustrate the above findings. The multilayered elastic medium is more complex in terms of the Rayleigh modes. Nevertheless, these new findings are still applicable for multilayered models.

DATA AVAILABILITY

The data underlying this paper will be shared on reasonable request to the corresponding author.

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